

Abstract

Driver fatigue and drowsiness represent a leading cause of commercial and passenger vehicle collisions worldwide, contributing to over 20% of fatal highway accidents. Conventional Driver Monitoring Systems (DMS) relying on cloud-based IoT architectures suffer from unpredictable network latency jitter (150 ms to 500+ ms), single points of failure in cellular dead zones (tunnels, mountain passes, and rural highways), prohibitive monthly data bandwidth expenses (\$15–\$30/month per vehicle), and severe biometric privacy liabilities under regulatory frameworks such as GDPR.

To overcome these fundamental limitations, this work presents **ALERT** (*Automated Lightweight Edge-based Real-time Tracking*), a 100% autonomous, privacy-preserving Non-IoT Edge driver monitoring system engineered specifically for sub-\$100 embedded hardware (Raspberry Pi 5 + Google Coral USB / Hailo-8L AI Accelerator) and high-performance automotive platforms (NVIDIA Jetson Orin Nano).

ALERT employs a non-invasive computer vision pipeline that extracts 468 3D facial landmarks to quantify key behavioral indicators: Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), PERCLOS (Percentage of Eye Closure over a 60-second moving window), and 3D Head Pose Euler angles (Pitch, Yaw, Roll). To resolve inter-person facial variation, the system incorporates an automated sub-10-second personalized baseline calibration ($EAR_{\text{threshold}} = \mu_{\text{EAR}} - 2.5 \cdot \sigma_{\text{EAR}}$) paired with exponential moving average drift adaptation and head-pose normalization.

For 24/7 operational robustness across harsh lighting conditions, ALERT integrates an active Near-Infrared (NIR) 850nm/940nm LED illumination array with narrow bandpass optical filters and adaptive CLAHE contrast enhancement (`clipLimit=4.0`). Eyewear and tinted sunglasses occlusions are addressed via a multi-layer degradation strategy that dynamically shifts monitoring weight to Mouth Aspect Ratio and downward head-nodding detection when eye confidence drops below threshold ($LCS < 0.40$).

To achieve ultra-low inference latency on constrained hardware, a 4-stage model compression methodology was implemented—combining MediaPipe BlazeFace and Face Mesh backbones, Knowledge Distillation from a ResNet-50 teacher network, 30% structured channel pruning, and Quantization-Aware Training (QAT INT8). Evaluated by a 3-Feature Support Vector Machine (SVM) classifier with an RBF kernel, ALERT achieves a **96.4% fatigue grading accuracy** at a deterministic **< 5 ms total pipeline latency** (enabling 200+ FPS capacity). By executing 100% on-device without cloud connectivity, ALERT guarantees instant life-saving alert generation, zero ongoing operational costs, and complete driver biometric data privacy.

Keywords: Driver Fatigue Detection, Edge AI, Computer Vision, Eye Aspect Ratio (EAR), PERCLOS, Model Compression, Quantization-Aware Training (QAT), Near-Infrared Vision, Raspberry Pi, Jetson Orin.